PROJECT PROMPT

Null Space, Range and Rank-Nullity Analyzer

Act as an experienced full-stack developer, Python programmer and Linear Algebra expert. I am a college student building a mini project for the subject Linear Algebra and Vector Calculus. Develop a complete, working and mathematically accurate application for the project below.

PROJECT

18. Null Space, Range and Rank-Nullity Analyzer

Modules: 1, 2, 3

AIM

Analyse matrices with different structural properties and verify the rank-nullity theorem.

INPUT

Several square and rectangular matrices.

- The user chooses the number of rows and columns, enters the entries (integers, negatives, fractions or decimals) and clicks Analyze.
- Validate the entries (no missing or invalid values).
- Include a few ready-made sample matrices so several square and rectangular cases can be tried quickly.

EXPECTED OUTPUT

For every matrix, display the following (show matrices and vectors in a clear matrix format):

1. RREF
2. Pivot structure
3. Rank
4. Nullity
5. Null-space basis
6. Range basis
7. Rank-nullity theorem verification

SEQUENTIAL MATHEMATICAL PROCESS

Matrix → Row reduction → Pivot analysis → Rank → Null space → Range → Rank-nullity verification

Implement and show the stages in exactly this order. Each stage must use the result of the previous one.

Stage 1 - Matrix
Read the entries and convert them into an m × n SymPy matrix A.

Stage 2 - Row reduction
Compute the RREF of A using exact SymPy arithmetic. Show A and its RREF.

Stage 3 - Pivot analysis
From the RREF, identify the pivot columns (leading variables) and the non-pivot columns (free variables). Use zero-based indexing internally but display column numbers from 1.

Stage 4 - Rank
Rank(A) = number of pivot columns.

Stage 5 - Null space
Solve Ax = 0 using the free variables. Show the general solution and a basis of the null space. Nullity(A) = number of basis vectors (= number of free variables). Verify that Av = 0 for every basis vector.

Stage 6 - Range
The basis of the column space is the set of pivot columns taken from the **ORIGINAL** matrix A, not from the RREF. Confirm that dim Range(A) = Rank(A).

Stage 7 - Rank-nullity verification
Check Rank(A) + Nullity(A) = n (number of columns), taking Rank from Stage 4 and Nullity from Stage 5 (do not compute nullity as n - rank). Show the substituted values (for example 2 + 1 = 3) and a clear "Verified" indicator.

MAJOR CONCEPTS TO DEMONSTRATE

RREF, rank, null space, range, basis, dimension, rank-nullity theorem.

SPECIAL CASES

Results must be correct for square, rectangular, zero, identity, singular, non-singular, full-rank and rank-deficient matrices, matrices with dependent columns, and matrices with fractional entries. Handle the trivial null space (only the zero vector: nullity 0, empty basis) and the zero matrix (rank 0, empty range basis) correctly.

TECHNOLOGY

Web application: Python (Flask) backend, HTML/CSS/JavaScript frontend, SymPy for all matrix calculations. It must run locally on Windows in VS Code. Keep the math logic in a separate module from the Flask routes. No databases, external APIs or unnecessary libraries.

WHAT TO GIVE ME

1. The project folder structure.
2. Complete working code for every file (no placeholders).
3. Step-by-step instructions to run it on Windows in VS Code.

Keep the code simple, modular and well-commented, suitable for a college mini-project. Prioritize mathematical correctness.